

Don't oversample the boundary: diagnose the minority deficit, then move the decision

Misclassified minority errors (mean over 79 datasets)



Most minority errors are threshold-recoverable: the classifier already ranks them correctly.

A threshold move reproduces oversampling's benefit — no synthetic data, better calibration.

Oversampling adds no ranking gain; at equal threshold tuning the generative vs. non-generative gap is within ± 1 pp.